Introduction

Hypertension (or high blood pressure) is a condition where the force that is caused by blood pushing along the artery walls is higher than usual and can lead to major heart complications and is categorized as a blood pressure reading of 130/80 mmHg or higher (*High Blood Pressure (Hypertension) - Symptoms & Causes - Mayo Clinic*, n.d.). A large portion of U.S. adults have hypertension with a reported 47.7% prevalence in 2023 (Fryar, 2024; *Products - Data Briefs - Number 511*, 2024). There are many risk factors for hypertension, including genetics/family history, race, older age, obesity, and lack of exercise (*High Blood Pressure (Hypertension) - Symptoms & Causes - Mayo Clinic*, n.d.). One of the potential complications that can come from hypertension is heart failure, as hypertension is one of the main risk factors (Reddy & Bergeron, 2025). Heart failure can occur when either side of the heart is weakened, leading to numerous complications and potentially death (*Heart Failure - Causes and Risk Factors | NHLBI, NIH*, 2022).

Digoxin is a medication that is meant to treat heart failure in combination with an angiotensin-converting enzyme (ACE) inhibitor and a diuretic and is in the digitalis glycosides class of medications (*Digoxin (Oral Route) - Side Effects & Dosage - Mayo Clinic*, n.d.). Associations between Digoxin use and reduced hospitalizations due to worsening heart failure in those with congestive heart failure (CHF) have been found (The Digitalis Investigation Group, 1997).

The purpose of this analysis is to explore whether the effect of Digoxin on hospitalizations due to worsening heart failure is modified by the history of hypertension reported at baseline among North American adults with CHF in the Dig clinical trial dataset. Hypertension as a major risk factor for CHF may interact with the effects of Digoxin on the prevention of worsening heart failure.

Methods

Survey/Dataset Description

The dataset for this secondary analysis regarding the potential effect modification of the previous history of hypertension on hospitalization due to worsening heart failure of the drug Digoxin comes from the Dig Trial. The data was collected and used to perform a large-scale clinical trial across over 300 centers in North America. The data contains information taken at baseline that consists of both demographic information (date of birth, sex, race) as well as health-related information that is associated with cardiac issues (history of hypertension, diabetes, ACE inhibitor type, ejection fraction, and more). The purpose of the original data collection was to test the effects of the drug Digoxin on outcomes related to heart failure in 6,800 subjects meeting the eligibility criteria of having heart failure and a left ventricular ejection fraction of 0.45 or less, primarily concerned with mortality related to cardiovascular issues. Baseline data were originally collected in 1991, and follow-up data were collected until mid-1995. Subjects were randomized to a Digoxin treatment group or a placebo group.

Statistical Methods

A stratified subgroup analysis exploring the effect of Digoxin vs placebo on a primary outcome of hospitalization due to worsening heart failure at a 2-sided significance level of 0.05 will be conducted, stratifying by previous instance of hypertension at baseline among all subjects included in the Dig dataset without missing data regarding treatment group, previous hypertension, and hospitalization due to worsening heart failure. If any of the selected baseline covariates including age at randomization, race, NYHA functional class, BMI, ejection fraction (percent), duration in months of CHF, and the number of symptoms of CHF chosen are not evenly distributed across treatment and placebo groups in each previous hypertension subgroup, they will be included as a factor for adjustment in a multivariable logistic regression model testing for the effect of treatment group (Digoxin vs placebo) on hospitalization due to worsening heart failure stratified by previous hypertension reported at baseline. The subgroup analysis will compare the calculated risk difference, risk ratio, and odds ratio, and their 95% confidence intervals, on the binary outcome of hospitalization due to worsening heart failure between the Digoxin and placebo groups. The potential interaction between previous hypertension and treatment assignment on hospitalization due to worsening heart failure will be statistically assessed via a Breslow-Day test for equal odds ratios at the 0.05 significance level. Statistical analysis and visualizations will be done in SAS Version 9.4.

Results

TABLE 1. STRATIFIED AND OVERALL FREQUENCIES AND RISK OF WHF

	Previous Hypertension		**No Previous Hypertension**		**Overall**	
n(%)	Digoxin	Placebo	Digoxin	Placebo	Digoxin	Placebo
Hospitalization due to WHF	422 (27.6%)	547 (35.1%)	488 (26.1%)	633 (34.3%)	910 (26.8%)	1180 (34.7%)
No hospitalization due to WHF	1105 (72.4%)	1010 (64.9%)	1381 (73.9%)	1213 (65.7%)	2486 (73.2%)	2223 (65.3%)

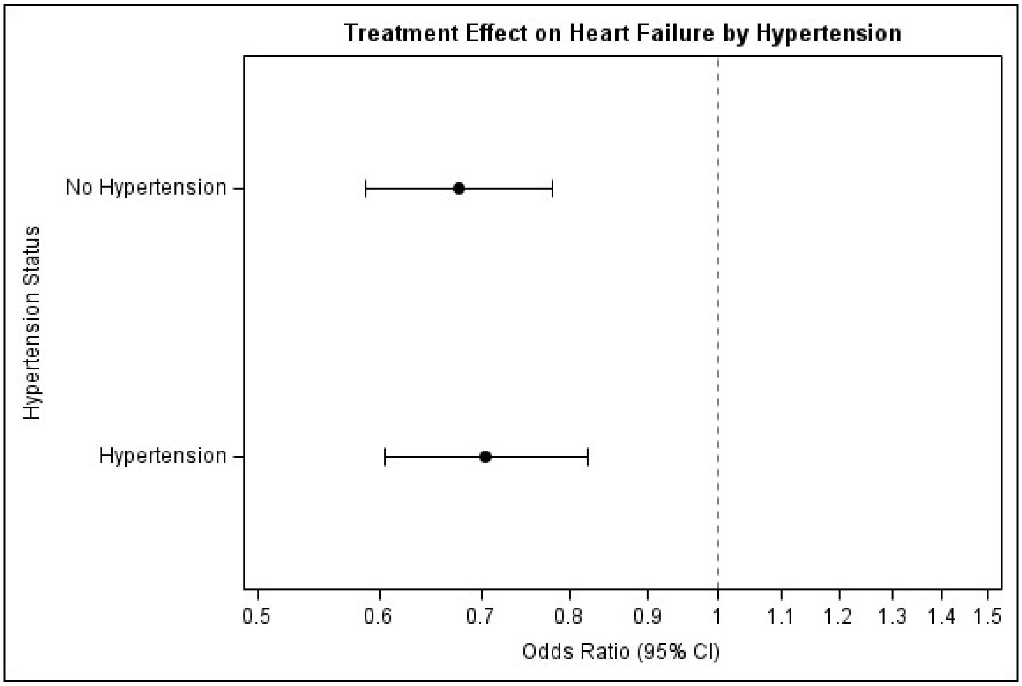
There were 3396 and 3404 subjects randomized into the Digoxin group and placebo group, respectively. Among the Digoxin group, 1527 reported previous hypertension at baseline, and 1869 reported no history of hypertension at baseline. 1557 subjects in the placebo group reported previous hypertension at baseline, and 1846 did not. Due to the randomized clinical trial study design, the subgroups were balanced for all other potential confounding variables, and no other adjustments were included in the analyses. The risk of hospitalization due to WHF remained consistent across experimental groups in the previous hypertension at baseline and no previous hypertension at baseline subgroups, with a 27.6% and 26.1% risk in those randomized to Digoxin, and 35.1% and 34.3% risk in those randomized to placebo, respectively. The overall risks in the Digoxin and placebo groups reflected the risks in the subgroups, with a 26.8% risk in the Digoxin arm and 34.7% risk in the placebo arm (Table 1).

TABLE 2. STRATIFIED AND OVERALL EFFECT OF DIGOXIN VS PLACEBO ON WHF

Effect of Digoxin vs Placebo on Hospitalization due to WHF	Previous Hypertension	No Previous Hypertension	Overall
Risk Difference (95% CI)	-0.08 (-0.11, -0.04)	-0.08 (-0.11, -0.05)	-0.08 (-11.0, -0.6)
Risk Ratio (95% CI)	0.79 (0.71, 0.87)	0.76 (0.69, 0.84)	0.77 (0.72, 0.83)
Odds Ratio (95% CI)	0.71 (0.61, 0.82)	0.68 (0.59, 0.78)	0.69 (0.62, 0.76)
Breslow-Day Test Results	Chi-Square: 0.146	P=0.703	DF: 1

The risk of hospitalization due to WHF was very similar between the Digoxin and placebo arms across both subgroups and the overall randomized study population. These remained consistent across both the absolute and relative difference scales, with similar risk differences, risk ratios, and odds ratios (and overlapping 95% confidence intervals). Table 2 depicts these differences, as well as the results from the Breslow-Day test of equal odds ratios between the subgroups. Subjects who reported previous hypertension at baseline had an odds ratio (95% CI) for hospitalization due to WHF of 0.71 (0.61, 0.82) comparing Digoxin to placebo arms, and subjects who reported no previous hypertension at baseline were found to have an odds ratio (95% CI) of 0.68 (0.59, 0.78). Both odds ratios and 95% confidence intervals suggest significant decreases in the odds of hospitalization due to WHF in the Digoxin arms compared to the placebo arms in both subgroups, consistent with the overall study population findings (odds ratio [95% CI] of 0.69 [0.62, 0.76]). Statistically assessing the interaction of previous hypertension reported at baseline via a Breslow-Day test at a 5% significance level further illustrated a lack of significant difference in the odds ratios (Chi-square=0.146, DF=1, p=0.703), which led to failing to reject the null hypothesis of no interaction. The subgroup odds ratios, 95% confidence intervals, and their overlapping nature are further illustrated in Figure 1.

FIGURE 1. STRATIFIED ODDS RATIOS OF DIGOXIN VS PLACEBO ON WHF



Discussion

The results from this exploratory analysis of the potential interacting effects that previous hypertension at baseline would have on the overall treatment effect of Digoxin versus placebo on hospitalization due to WHF found no significant effect modification. The odds of the outcome of interest occurring were not significantly different across both subgroups and reflected the overall pooled study odds as well. Among the study population of U.S. adults who met the criteria for having experienced heart failure, Digoxin use led to decreased risks and odds for hospitalization due to WHF compared to placebo in both those who reported previous hypertension at baseline and those who did not, with consistent treatment effects across the subgroups.

Hypertension remains a major risk factor for WHF, and potential interactions and effects between the two should be further explored (Reddy & Bergeron, 2025). Links between sex, age, race, and gender to increased risk of WHF have been found, and further studies into these discrepancies should be explored as well (Rossello et al., 2021). Some limitations of this report include the lack of exploration into further subgroups and the effect of Digoxin on hospitalization due to WHF, as well as the choice to explore a potential link between WHF and a known risk factor, not necessarily fully coinciding with the mechanism of Digoxin itself. Further research exploring the mechanisms of Digoxin and whether these biologically differ between subsets of individuals with a greater risk of WHF or any cardiac-related outcome of interest would be beneficial.

References

Digoxin (oral route)—Side effects & dosage—Mayo Clinic. (n.d.). https://www.mayoclinic.org/drugs-supplements/digoxin-oral-route/description/drg-20072646

Heart Failure—Causes and Risk Factors | NHLBI, NIH. (2022, March 24). https://www.nhlbi.nih.gov/health/heart-failure/causes

High blood pressure (hypertension)—Symptoms & causes—Mayo Clinic. (n.d.). https://www.mayoclinic.org/diseases-conditions/high-blood-pressure/symptoms-causes/syc-20373410

The Digitalis Investigation Group, (1997). The Effect of Digoxin on Mortality and Morbidity in Patients with Heart Failure. *New England Journal of Medicine*, *336*(8), 525–533. https://doi.org/10.1056/NEJM199702203360801

Products—Data Briefs—Number 511 -October 2024. (2024, November 5). https://doi.org/10.15620/cdc/164016

Reddy, Y. N. V., & Bergeron, N. (2025). Risk Factors for Contemporary Heart Failure—Respecting the Old (Hypertension) While Accepting the New (Obesity). *Mayo Clinic Proceedings*, *100*(8), 1275–1277. https://doi.org/10.1016/j.mayocp.2025.06.010

Rossello, X., Ferreira, J. P., Caimari, F., Lamiral, Z., Sharma, A., Mehta, C., Bakris, G., Cannon, C. P., White, W. B., & Zannad, F. (2021). Influence of sex, age and race on coronary and heart failure events in patients with diabetes and post-acute coronary syndrome. *Clinical Research in Cardiology*, *110*(10), 1612–1624. https://doi.org/10.1007/s00392-021-01859-2